

Thamilmani Murugan

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Summary

Experienced and passionate CFD researcher specializing in **turbulence, heat transfer, energy systems, aerodynamics**, and turbo-machinery. PhD. thesis submitted at **IIT Bombay** focusing on **supercritical heat transfer**. Proven track record of tackling complex engineering challenges, with active alignment toward dynamic job opportunities to drive technical learning, contribution, and professional growth.

Areas of Research

Turbulence Modeling, Supercritical heat transfer, CFD based Optimization, Hydraulic and Thermodynamic System analysis, Aerodynamics.

Skills

- **Programming & Scripting Languages:** Python, C, C++, MATLAB, SciLab, BASH
- **Tools & Libraries:** ANSYS Fluent, CFX, OpenFOAM, AutoCAD, SOLIDWORKS

Work Experience

Research Consultant | *HINCOL India - IITB Collaborative Project* (Jan'26-Sep'26)

- Audited the existing tank design and the plant layout for modified bitumen production plant.
- Developed theoretical frameworks for designing new tanks, thermal losses and hydraulic calculations for **decreasing the modified bitumen production time within 2 hours**.
- Performed a system-wide **pressure and thermal loss calculations** to assist in heater sizing and pump sizing.

Research Consultant | *Tabrej DesignLife, Thane* (Dec'22-Feb'23)

- Conducted a **feasibility study** and Thermodynamic system analysis of a proposed HVAC System.

Project Officer | *Center of Propulsion Technology, IIT Madras, Chennai* (July'18-Dec'18)

- Conducted numerical studies on the "Unsteady flow physics and **loss mechanisms** and shock boundary layer interactions in turbine flow field" under Prof. BVSSS Prasad for DRDO Project using ANSYS CFX.

Education

PhD. Mechanical Engineering | *IIT Bombay* | **7.71 CPI** (Jan'19-May'26)

- **Thesis:** *Characterization, modeling and mitigation of heat transfer deterioration in supercritical fluids.*
- Developed a novel **turbulent Prandtl number model** and Nusselt-number correlations, achieving < 10% error in heat transfer predictions compared to experimental benchmarks for supercritical fluids.
- Investigated pulsations to disrupt thermal boundary layers to mitigate the heat transfer deterioration process, employing **Dynamic Mode Decomposition (DMD)** to isolate coherent spatial structures and critical frequencies.
- Executed **CFD based optimization** for the selection of baffle size for effective heat transfer enhancement under supercritical conditions.

M.Tech. Thermal & Fluids Engineering | *Amrita University, Kerala* | **9.48 CGPA Gold Medalist** (Aug'16-June'18)

- **Thesis:** *Numerical analysis of flow over multiple square cylinders in tandem arrangement*
- Numerically investigated flow interactions between multiple square cylinders in tandem to assess the criticality of **drag reduction behavior** in the wake region.
- Served as a Project Intern at IIT Bombay at PIV Lab, under Prof. Amit Agrawal.

- **Thesis:** *Studies on inward solidification of PCM dispersed with macro-particles.*
- Analysed the inward **solidification of phase change material** with macro-particles to assess the effect of particle fraction and thermal conductivity on solidification time using in-house C code.

Self Projects

- Collaborated with MS. Research student from Western University, Canada for implementing Reduced order model for **moving boundary method in Heat pipes**.
- **foam_to_fluent2D:** Built Python based custom automation workflow to convert 2.5D OpenFOAM Meshes to 2D ANSYS Fluent compatible meshes.
- **residual_HPC:** Built Python based CLI tool for simultaneous residual monitoring of multiple remote ANSYS Fluent jobs that are run in ROCKS or OpenHPC Clusters.

Journal Publications

- Murugan *et al.*, "Numerical Analysis of Pulsatile Flow of Supercritical Fluids Through a Vertical Pipe", *Heat Transfer Journal*, Oct 2026. (DOI: 10.1002/htj.70399)
- Murugan *et al.*, "Proposal of turbulent Prandtl number models for predicting heat transfer deterioration in supercritical flows," *Physics of Fluids*, Sep 2025. (DOI: 10.1063/5.0284649)
- Murugan *et al.*, "Study on Convective Heat Transfer of Supercritical Water in Annular Square Channel", *Heat Transfer Engineering*, Jan 2023. (DOI: 10.1080/01457632.2022.2162012)
- Narasimhan *et al.*, "Studies on the inward spherical solidification of a phase change material dispersed with macro particles", *Journal of Energy Storage*, Apr 2018. (DOI:10.1016/j.est.2017.10.016)

Conference Presentations

- "Characterization of Helmholtz oscillator for mitigating heat transfer deterioration in supercritical flows," **ECOS 2025**, Ecole des Mines, PSL, Paris, France, June 29 – July 4, 2025.
- "Numerical studies on flow over multiple cylinders in tandem arrangement: Effect of number of cylinders," **FMFP 2018**, IIT Bombay, Mumbai, India, Dec 2018.

Teaching Assistance

- **ME684** : Air Conditioning system design (Jan'20-Apr'20)
- **ME661** : Advanced Thermodynamics and Combustion (Aug'20-Dec'20 & Aug'21-Dec'21)
- **ME415** : Computational Fluid dynamics and Heat transfer (Jan'21-Apr'21 & Jan'22-Apr'22)
- **ME346** : Heat transfer (Aug'22-Dec'22)
- **ME306** : Applied Thermodynamics (Jan'23-Apr'23)

POR & Extra-curricular Activities

- Organized and **Co-hosted** ASME India Seminar on "Engineering Education with focus on Computational Mechanics", held at IIT Bombay on August 9, 2025.
- Managed and maintained HPC, **ROCKS and OpenHPC based cluster** for the Thermal Science Lab under Prof. Sandip Kumar Saha during 2024-26.
- Organized events **leading more than 50 member team** for Tamil literary club "Saaral" for Annual cultural festival "Instincts" in 2015.
- Core Team Member of "Teach a School" Project, with SSN Lakshya to lead volunteers to impart science-based learning to school children in 2015.
- Member of **SAE SUPRA F2 Team**, "Mach Racing Team" from SSN College of Engineering, for the year 2014.